A UNIFORM CLOSED FORM FOR MODULAR SCHUR NUMBERS $S_m(k,\ell)$

ADAM MCKENNA

Abstract. For integers $m \geq 2$, $\ell \geq 2$, and $k \geq 1$, the *modular Schur number* $S_m(k,\ell)$ of D'orville–Sim–Wong–Ho [1] is the greatest N such that $[1,N]$ admits a k-partition into classes that are ℓ-sum-free modulo m. We prove the closed form

$$S_m(k,\ell) \;=\; \frac{m}{\gcd(m,\ell-1)} - 1$$

for all $m \geq 2$, $\ell \geq 2$, and $k \geq m/\gcd(m,\ell-1)-1$. This identifies the stabilized value of $S_m(k,\ell)$ for every modulus and settles the plateau phenomenon observed empirically in [1] for $m \leq 7$. The main theorem has been formalized and machine-verified in Lean 4. We also record a sharp threshold result for prime moduli ($k_0(p,\ell) = p-1$ for $\ell \geq p-1$, $\ell \not\equiv 1 \pmod p$) and refute a natural singletons-only conjecture in the composite case via the explicit witness $\{1,5\} \subseteq \mathbb{Z}/12$.

1. Introduction

1.1. **Background.** The classical Schur number $S(k)$ is the largest integer N such that $\{1,\dots,N\}$ admits a k-partition into sum-free sets. Values are known only through $S(5) = 160\,537$ (Heule, 2017); computing $S(6)$ is the content of Erdős problem 483 and remains open. D'orville, Sim, Wong, and Ho [1] introduced the natural modular generalization:

Definition 1.1. A set $S \subseteq \mathbb{Z}$ is *ℓ-sum-free modulo m* if no $x_1,\dots,x_\ell \in S$ (not necessarily distinct) satisfy $x_1+\cdots+x_\ell \equiv y \pmod m$ for any $y \in S$. The *modular Schur number* $S_m(k,\ell)$ is the greatest $N \geq 0$ such that $[1,N]$ admits a k-partition into ℓ-sum-free sets modulo m.

In [1] complete formulas are given for $m \in \{1,2,3,4,5,6,7\}$ via explicit residue-by-residue case analysis. The phenomenon visible in those formulas—once k is large enough, $S_m(k,\ell)$ is a function of $\gcd(m,\ell-1)$ only—was conjectural for general m.

1.2. **Main result.** Let $d := \gcd(m,\ell-1)$ and $n := m/d$. Our main theorem makes the plateau explicit and dispenses with any hypothesis on m.

Theorem 1.2 (Unified Stabilized Law)**.** *For all integers $m \geq 2$, $\ell \geq 2$, and $k \geq n-1$,*

$$S_m(k,\ell) \;=\; n-1 \;=\; \frac{m}{\gcd(m,\ell-1)} - 1.$$

This recovers the stabilized portions of the $m \leq 7$ tables of [1] as corollaries (§6), extends them to all m, and resolves the plateau conjecture.

1.3. **Formalization.** The proof is short and elementary, but the authors have found its unified form easy to misstate (earlier drafts confused the role of n versus m in the forbidden tuple). To eliminate doubt, Theorem 1.2 has been formalized in Lean 4 and machine-verified by Aristotle [3]. The theorem statement

```
theorem schurModResidue_eq (m k l : N) (hm : 2 <= m) (hl : 2 <= l)
    (hk : m / Nat.gcd m (l - 1) - 1 <= k) :
    schurModResidue m k l = m / Nat.gcd m (l - 1) - 1
```

compiles in Lean 4 with axiom dependency only `propext`, `Classical.choice`, `Quot.sound`. Source files are available at `https://github.com/mysticflounder/modular-schur/tree/main/lean`.

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1.4. **Organization.** §2 establishes base lemmas. §3 proves the upper bound $S_m(k,\ell) \le n-1$. §4 proves the matching lower bound. §5 analyzes the threshold $k_0(m,\ell)$ and records tight prime-modulus results. §6 lists corollaries for specific m. §7 poses open questions on tight k_0 for composite m.

2. Preliminaries

Throughout, $m \ge 2$ and $\ell \ge 2$ are fixed integers, $d := \gcd(m, \ell-1)$, and $n := m/d$. Note $dn = m$ exactly, and $d \mid (\ell-1)$ by definition.

Lemma 2.1 (Residue reduction)**.** *If $N \le m-1$, the integers $1, 2, \ldots, N$ have pairwise distinct nonzero residues modulo m. A partition of $[1, N]$ into classes $C_1 \sqcup \cdots \sqcup C_k$ is valid if and only if each C_i, viewed as a subset of $\{1, \ldots, m-1\} \subseteq \mathbb{Z}/\mathrm{m}$, is ℓ-sum-free modulo m.*

Proof. All elements of $[1, N]$ lie in $[1, m-1]$, and the map $[1, m-1] \to \mathbb{Z}/\mathrm{m}$ is injective onto $\{1, \ldots, m-1\}$. Sum-freeness is preserved under this identification. □

Lemma 2.2 (Universal upper bound)**.** *For every $\ell \ge 1$ and every $k \ge 1$, $S_m(k,\ell) \le m-1$.*

Proof. If $N \ge m$, then $m \in [1, N]$ lies in some class C. The tuple $(m, \ldots, m; m) \in C^{\ell+1}$ satisfies $m \cdot \ell \equiv 0 \equiv m \pmod{m}$, violating ℓ-sum-freeness. □

Lemma 2.3 (Singleton safety)**.** *For $r \in \{1, \ldots, m-1\}$, the singleton $\{r\} \subseteq \mathbb{Z}/\mathrm{m}$ is ℓ-sum-free modulo m if and only if $(\ell-1)r \not\equiv 0 \pmod{m}$; equivalently, if and only if $n \nmid r$.*

Proof. $\ell\{r\} \cap \{r\} \ne \emptyset$ iff $\ell r \equiv r \pmod{m}$ iff $(\ell-1)r \equiv 0 \pmod{m}$. Write $\ell - 1 = du$ so $\gcd(u, n) = 1$; then $(\ell-1)r \equiv 0 \pmod{dn}$ iff $ur \equiv 0 \pmod{n}$ iff $n \mid r$. □

Lemma 2.4 (Monotonicity in k)**.** *If $[1, N]$ admits a valid k_0-partition, then for every $k \ge k_0$ it admits a valid k-partition.*

Proof. Pad with $k - k_0$ empty classes; the empty set is vacuously ℓ-sum-free. □

3. Upper bound

Theorem 3.1. *For all $m \ge 2$, $\ell \ge 2$, and $k \ge 1$,*

$$S_m(k,\ell) \;\le\; n-1.$$

Proof. Suppose $N \ge n$. Since $1 \le n \le m$, we have $n \in [1, N]$; let C be the class of a valid k-partition containing n. Write $\ell - 1 = du$ with $u \in \mathbb{Z}$, so

$$(\ell-1) \cdot n \;=\; du \cdot n \;=\; u \cdot (dn) \;=\; u \cdot m \;\equiv\; 0 \pmod{m}.$$

Hence $\ell n \equiv n \pmod{m}$. The tuple $(n, \ldots, n; n) \in C^{\ell+1}$ thus witnesses ℓ-sum-freeness failure in C, a contradiction. □

Remark 3.2. The argument unifies two apparently distinct obstructions: the "$N \ge m$ is blocked by m itself" lemma (Lemma 2.2, the $d = 1$ case where $n = m$) and the "$r = m/d$ is an unsafe singleton" lemma (Lemma 2.3 at $r = n$). In both cases the obstruction is the single element $n \in [1, N]$ being ℓ-sum-free-unsafe as a singleton; once $N \ge n$, this element lands in *some* class, killing the partition.

4. Lower bound

Theorem 4.1. *For all $m \geq 2$, $\ell \geq 2$, and $k \geq n-1$,*

$$S_m(k,\ell) \;\geq\; n-1.$$

Proof. If $n = 1$ the conclusion reads $S_m(k,\ell) \geq 0$, vacuously true. Assume $n \geq 2$. Partition $[1, n-1]$ into singletons $D_r := \{r\}$ for $r = 1, \ldots, n-1$. By Lemma 2.3 each D_r is ℓ-sum-free (since $1 \leq r < n$ forces $n \nmid r$). This is a valid $(n-1)$-partition; for $k \geq n-1$ pad with empty classes by Lemma 2.4. □

Proof of Theorem 1.2. Combine Theorems 3.1 and 4.1. □

5. The threshold $k_0(m,\ell)$

Let $k_0(m,\ell)$ denote the least k such that $S_m(k',\ell) = n-1$ for every $k' \geq k$.

Theorem 5.1. $k_0(m,\ell) \leq \max(1,\, n-1)$.

Proof. Theorem 4.1 exhibits a valid $(n-1)$-partition achieving $N = n-1$ when $n \geq 2$. When $n = 1$ the value is zero, achieved trivially at $k = 1$. □

Theorem 5.2 (Covering lower bound)**.** *Let $\sigma(m,\ell) := \max\{|C| : C \subseteq \{1,\ldots,n-1\},\ C$ is ℓ-sum-free mod $m\}$. Then $k_0(m,\ell) \geq \lceil (n-1)/\sigma(m,\ell) \rceil$.*

Proof. If $[1, n-1]$ admits a valid k-partition, summing class sizes gives $n-1 \leq k \cdot \sigma(m,\ell)$. □

5.1. Tight threshold for prime modulus.

Theorem 5.3 (Large-ℓ coset criterion)**.** *Let $C \subseteq \mathbb{Z}/\mathrm{m}$ with $|C| \geq 2$, pick $a_0 \in C$, and let $H := \langle C - C \rangle = g\,\mathbb{Z}/\mathrm{m}$ with $g := m/|H|$. For every $\ell \geq |H| - 1$,*

$$\ell C \;=\; \ell a_0 + H,$$

and consequently C is ℓ-sum-free modulo m if and only if $g \nmid (\ell-1)a_0$.

Proof sketch. ($\subseteq$) Every ℓ-sum $\sum c_i = \ell a_0 + \sum (c_i - a_0)$ lies in $\ell a_0 + H$ since $c_i - a_0 \in H$.

($\supseteq$) Pick $a_1 \in C$ with $\langle a_1 - a_0 \rangle = H$ (such a_1 exists because differences generate H, and in a cyclic group some single difference generates the full subgroup generated by a finite set). Writing $\delta := a_1 - a_0$, the sequence $0, \delta, 2\delta, \ldots, (|H|-1)\delta$ exhausts H, so for any $h \in H$ there is $j \in \{0, \ldots, |H|-1\}$ with $h = j\delta$; take $c_1 = \cdots = c_j = a_1$ and $c_{j+1} = \cdots = c_\ell = a_0$ (valid since $j \leq |H| - 1 \leq \ell$) to realize $\ell a_0 + h$ as an ℓ-sum.

The safety criterion follows: since $C \subseteq a_0 + H$, we have $\ell C \cap C \neq \emptyset$ iff $\ell a_0 + H = a_0 + H$ iff $(\ell-1)a_0 \in H$ iff $g \mid (\ell-1)a_0$. □

Corollary 5.4 (Sharp k_0 for prime moduli)**.** *Let p be prime. For every $\ell \geq p-1$ with $\ell \not\equiv 1 \pmod{p}$,*

$$k_0(p,\ell) \;=\; p-1.$$

Proof. For any $|C| \geq 2$ in $\mathbb{Z}/\mathrm{m}[p]$, the subgroup $\langle C - C \rangle$ is either 0 (ruled out) or $\mathbb{Z}/\mathrm{m}[p]$ (since $\mathbb{Z}/\mathrm{m}[p]$ has no nontrivial subgroups). By Theorem 5.3 with $g = 1$, $\ell C = \mathbb{Z}/\mathrm{m}[p]$, so C is not ℓ-sum-free. Hence $\sigma(p,\ell) = 1$, and Theorem 5.2 gives $k_0 \geq n-1 = p-1$; Theorem 5.1 supplies the match. □

5.2. **Composite moduli: refuting a natural conjecture.** One might conjecture that for every m and all sufficiently large ℓ with $\gcd(m, \ell - 1) < m$, only singletons are ℓ-sum-free, so $k_0(m, \ell) = n - 1$. This is false in the composite case.

Proposition 5.5. *For $m = 12$ and every $\ell \equiv 11 \pmod{12}$ (so $d = 2$, $n = 6$), the set $C = \{1, 5\} \subseteq \{1, \ldots, 5\}$ is ℓ-sum-free modulo 12. Consequently $\sigma(12, \ell) \geq 2$ and $k_0(12, \ell) \leq 3 < n - 1$.*

Proof. $C - C = \{0, \pm 4\}$, so $\langle C - C \rangle = 4\mathbb{Z}/12 = \{0, 4, 8\}$, $|H| = 3$, $g = 4$. Theorem 5.3 applies once $\ell \geq 2$. With $a_0 = 1$, the criterion $4 \nmid (\ell - 1) \cdot 1$ reads $4 \nmid \ell - 1$, i.e. $\ell \not\equiv 1 \pmod 4$. For $\ell \equiv 11 \pmod{12}$ we have $\ell \equiv 3 \pmod 4$, so $4 \nmid \ell - 1$, and C is ℓ-sum-free. Combined with singletons $\{2\}, \{3\}, \{4\}$ this gives a valid 3-partition of $[1, 5]$. □

6. Corollaries: recovered and extended formulas

Specializing Theorem 1.2:

- $m = 8$**:** for $k \geq 7$, $S_8(k, \ell)$ equals $0, 1, 3, 7$ according to $\gcd(8, \ell - 1) \in \{8, 4, 2, 1\}$ (residue pattern $\ell \bmod 8$).
- $m = 9$**:** for $k \geq 8$, $S_9(k, \ell)$ equals $0, 2, 8$ according to $\gcd(9, \ell - 1) \in \{9, 3, 1\}$.
- $m = 11$**:** for $k \geq 10$, $S_{11}(k, \ell) = 0$ if $\ell \equiv 1 \pmod{11}$, else 10.
- **Prime-power** $m = p^e$**:** writing $v := \min(v_p(\ell - 1), e)$, for $k \geq p^{e-v} - 1$,

$$S_{p^e}(k, \ell) \;=\; p^{e-v} - 1.$$

All of these stabilized rows are consistent with (and for $m = 4, 5, 6, 7$ recover) the tables of [1, §2–3].

7. Open problems

(1) *Tight k_0 for composite m.* Theorem 5.1 gives $k_0 \leq n - 1$ and Theorem 5.2 gives $k_0 \geq \lceil (n - 1)/\sigma(m, \ell) \rceil$. These match in the prime case (Corollary 5.4) but not generally: for $m = 8$ with ℓ even, $\sigma(8, \ell) = 4$ (witness $\{1, 3, 5, 7\}$), giving only $k_0 \geq 2$, whereas the true value is $k_0 = 3$. A closed form for $k_0(m, \ell)$ when m is composite remains open.

(2) *Boundary regime.* For $1 \leq k < n - 1$, $S_m(k, \ell)$ is not controlled by Theorem 1.2; empirical data through $m \leq 13$ suggest the values are eventually periodic in $\ell \bmod m$ for each fixed k, but no closed form is known.

(3) *Asymptotics as $m \to \infty$.* What is $\max_{\ell \geq 2} S_m(k, \ell)$ as a function of m for fixed k? The unified law gives the pointwise value; the optimization over ℓ is a divisor-counting problem.

References

[1] R. D'orville, C. Sim, E. Wong, and T.-K. L. Ho, *Modular generalizations of Schur numbers*, Integers **25** (2025), #A62. `https://math.colgate.edu/~integers/z62/z62.pdf`

[2] M. J. H. Heule, *Schur number five*, AAAI 2018, 6598–6606. `arXiv:1711.08076`.

[3] T. Achim *et al.*, *Aristotle: IMO-level Automated Theorem Proving*, 2025. `arXiv:2510.01346`.